

NLP Project Proposal - Optimal Quantization in MoE

Post-training quantization reduces LLM memory, but it introduces severe **routing instability** in Mixture-of-Experts (MoE) architectures by scrambling intermediate gating logits and causing expert choices to degrade. While existing literature tries to fix this via complex mathematical calibrations during compression, we want to explore a much simpler structural solution. Because router layers make up a microscopic fraction of the model's total parameters, we ask: **Can we bypass expensive mathematical optimizations by keeping only the tiny routing layers in high precision while uniformly quantizing 100% of the massive expert weights?** This mixed-precision setup aims to stop routing errors naturally with zero optimization overhead, directly testing an active debate regarding whether shielding router precision is sufficient to halt error propagation.

In order to do that we will compare three setups: the **original full-precision model**, a **fully quantized version**, and our **mixed-precision variant** where the router layers are protected.

We will use the lightweight *Qwen/Qwen2.5-MoE-1.4B-A14B* model evaluated on standard text benchmarks (WikiText-2/C4). Using PyTorch forward hooks attached to the gating modules, we will extract intermediate logits layer-by-layer to measure precise structural drift using *Kullback-Leibler (KL) Divergence*, *Top-k expert mismatches*, and *downstream perplexity*.

If the quantized experts output too much noise, the data may become too distorted for even a full-precision router to work. If this happens, we will try to test a variant that dynamically scales routing scores based on activation size to balance the noise. We will also track expert selection balance to see if quantization causes "expert collapse" (lazily dumping tasks onto a few popular experts) and verify if our safeguard keeps the rare, specialized experts working.

Regarding compute and storage, we would like to perform the experiments on the university's GPU cluster.

The relevant references:

1. Fu et al. (2025). *"EAQuant: Enhancing Post-Training Quantization for MoE Models."*
2. Anonymous (2026). *"Beyond Freezing the Router: Rank-Aligned Post-Training Quantization for MoE Models."*
3. Park et al. (2026). *"Value-and-Structure Alignment for Routing-Consistent Quantization of MoE Models."*

Hope you find it interesting,

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